

ML Challenge 2025: Smart Product Pricing

Solution Report

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1. Executive Summary

We build a **rule-compliant** text+structured pipeline for price prediction from `catalog_content`. Text is represented via TF-IDF (1–2 grams, 30k vocab) followed by **TruncatedSVD** (100 comps). Regex-derived quantity/format cues (e.g., *pack of*, numeric value hints, length/ratio stats) form structured features. The concatenated representation is fed to a **LightGBM** regressor. Model selection uses 5-fold CV with SMAPE.

2. Methodology Overview

2.1 Problem Analysis

Supervised regression from catalog descriptions; quantity and packaging indicators (e.g., “Pack of 2”, units like ml/g/kg/L) and text statistics correlate with price.

2.2 Approach

- **Text features:** `TfidfVectorizer` with `max_features = 30,000`, n-gram range (1,2), English stopwords; dimensionality reduced with **TruncatedSVD** to 100 components.
- **Structured features (from text):** `pack_of`, `value_num` (unit-aware numeric extract), `text_len`, `digit_ratio`, `upper_ratio`, `word_count`. Numerical features are imputed (0) and standardized with `StandardScaler`.
- **Fusion & model:** Concatenate R^{100} SVD features with 6 standardized numeric features $\rightarrow$ `LightGBMRegressor`.
- **Validation:** `KFold(n_splits=5, shuffle=True, random_state=42)` with SMAPE as the selection metric.

3. Model Architecture

3.1 Diagram

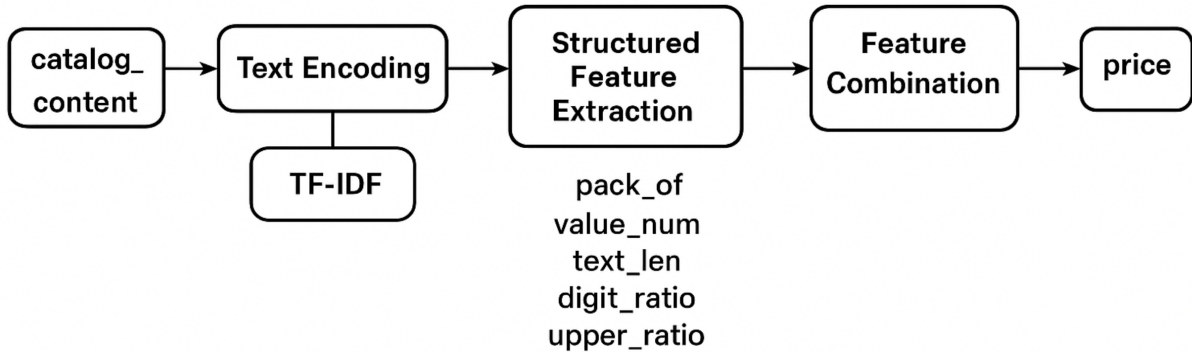


Figure 1: Pipeline — regex feature extractor, TF-IDF (1-2g, 30k) + SVD(100), feature scaling, concatenation, LightGBM regressor.

3.2 Key Implementation Details

- **Libraries:** scikit-learn (TF-IDF, SVD, scaling, KFold), LightGBM (lgbm).
- **LightGBM hyperparameters:** `n_estimators = 700`, `learning_rate = 0.05`, `num_leaves = 64`, `subsample = 0.8`, `colsample_bytree = 0.8`, `random_state = 42`, `n_jobs = 4`.
- **Training loop:** Fit on each fold, predict validation fold, compute SMAPE, and aggregate mean/std across folds.
- **Test-time:** Refit on full training data and predict on test.

4. Dataset, I/O and Compliance

4.1 Dataset Columns

- **id** (optional; created as $0..N-1$ if missing at test time), **catalog_content** (string), **price** (train only).

4.2 Output Format

CSV with exactly: `id`, `price`. One prediction is produced per test row. The submission file written by the code is `test.csv`.

4.3 Rule Compliance

- No external price lookup or internet augmentation.
- Only `catalog_content`-derived features are used (text + regex-engineered numerics).
- No image usage; no third-party API calls at train/inference time.

5. Training Objective and Metric

Model objective (fit): LightGBM default regression objective (squared error) on raw prices.

Model selection (validation): SMAPE:

$$\text{SMAPE} = 100 \times \frac{1}{N} \sum_i \frac{|\hat{y}_i - y_i|}{(|\hat{y}_i| + |y_i|) / 2}.$$

6. Results

6.1 Cross-Validation (5-fold)

Fold	SMAPE (%)	Notes
1	62.4860	TF-IDF+SVD(100) + 6 numeric feats
2	62.0528	
3	62.8194	
4	61.5835	
5	62.5711	
Mean	62.3026	Std: 0.4364

LightGBM reports using 106 features after concatenation in each fold.

6.2 Submission

The model was refit on the full training set and predictions were generated for the test set; the file `test.csv` was written successfully with columns `id`, `price`.

7. Conclusion

A compact **TF-IDF + SVD + structured-features** pipeline with a **LightGBM** regressor provides an efficient, fully rule-compliant baseline for smart product pricing. It captures key quantity/format signals from text and delivers stable validation performance under 5-fold SMAPE evaluation.